

Javascript

Loops, jump statements, dialogue box , function

```
<!doctype html>
<html lang="en">
  <head>
    <meta charset="UTF-8" />
    <meta name="viewport" content="width=device-width, initial-scale=1.0" />
    <title>Hoisting</title>
  </head>
  <body>
    <h1>Loops</h1>

    <!-- for , while, do-while -->

    <script>

      // funtion -> block of code

      function addition(a, b){ // parameter
        console.log(a+b);
      }

      // run a function
      addition(10, 30); // argument
      addition(90, 70);

      function multiply(a,b, c=30){
        console.log(a*b*c);
      }

      multiply(10,40);

      // let result = Number(prompt("Enter your age"));

      // console.log(typeof result);

      // if(result >= 18){
      //   alert("You are eligible for vote");
      // }else{
      //   alert("You are not eligible for vote");
      // }
```

```

    // let result = confirm("Are you sure to delete this file");
    // if(result){
    //     alert("Your file is deleted successfully")
    // }else{
    //     alert("your file is safe")
    // }
    // console.log(result);

console.log("loops");

for (let i = 1; i <=5; i++) {
    // if(i == 15){
    //     break;
    // }
    // alert(i);

    // if(i>=5 && i<=10){
    //     continue;
    // }
    console.log(i);
}

// for(let i=10; i>0; i--){
//     console.log(i);

// }

// for(let i=1; i<=5; i++){

//     for(let j=1; j<=i; j++){

//         console.log("* ");

//     }
//     console.log("\n");

// }

// let j = 0;
// while (j <= 20) {
//     document.write(j);
//     j++;
// }

// let k=10;

// do{

```

```
//      document.write(k, " ");  
//      k++;  
//    }while(k<10);  
</script>  
</body>  
</html>
```

Theory Notes

1. Functions

- **Definition:** Block of code that performs a task.
- **Parameters vs Arguments:**
 - Parameters → placeholders in function definition.
 - Arguments → actual values passed when calling the function.
- **Default parameters** → function multiply(a, b, c=30) assigns a default value if not provided.

2. Loops

- **for loop** → runs a block of code a fixed number of times.
- **while loop** → runs until condition becomes false.
- **do-while loop** → runs at least once, then checks condition.
- **Nested loops** → loop inside another loop (used for patterns, matrices).
- **Control statements:**
 - break → exits loop immediately.
 - continue → skips current iteration and moves to next.

3. User Interaction

- prompt() → input from user.
- alert() → show message box.
- confirm() → yes/no dialog returning true or false.

Coding Round Questions

1. Write a function sum(a, b) that returns the sum of two numbers.
2. Create a function with a default parameter and demonstrate calling it with and without the third argument.
3. Use a for loop to print numbers from 1 to 10.

4. Write a nested loop to print a triangle pattern of *.
5. Demonstrate break and continue inside a loop.
6. Use prompt() to ask the user's age and print if they are eligible to vote.
7. Use confirm() to ask if the user wants to delete a file and print the result.

Machine Coding Round Challenge

Problem: Simple ATM Simulation

Requirements:

- Create a function atm() that:
 - Asks user for a PIN using prompt().
 - If PIN is correct, show a menu with options:
 1. Check Balance
 2. Withdraw
 3. Deposit
 4. Exit
 - Use loops to keep showing the menu until user chooses Exit.
 - Use conditional statements to handle each option.
- Maintain balance in a variable and update it accordingly.